EDS 223 HW #1

Lead Paint Risk in Philadelphia for Communities of Color

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Mapping disproportionate levels of lead paint risk in Philadelphia County

With its old buildings and legacy environmental contamination, Philadelphia has one of the highest risks for lead exposure of any U.S. city. Additionally, research has found lead to disproportionately affect people of color and poorer communities. This analysis will focus on how lead paint risk factors relate to people of color in Philadelphia.

Objective: Visualize the spatial correlation between percent lead paint risk and percent people of color across Philadelphia County Census Blocks.

Load packages

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```
library(sf)
```

```
Linking to GEOS 3.14.1, GDAL 3.12.1, PROJ 9.7.1; sf_use_s2() is TRUE
```

```
library(here)
```

```
here() starts at C:/Users/sabus/localeds/Git/MEDS/eds223/EDS223-HW1
```

```
library(tmap)
```

```
Registered S3 method overwritten by 'stars':
  method                  from
  st_interpolate_aw.stars sf
```

```
library(stars)
```

```
Loading required package: abind
```

Import data

The Census block-level data from the US EPA's EJScreen is read.

```
# Read in geodatabase of EJScreen data at the Census Block Group level
ejscreen <- sf::st_read(here::here("data", "ejscreen","EJSCREEN_2023_BG_StatePct_with_AS_CNM
```

```
Reading layer `EJSCREEN_StatePctiles_with_AS_CNMI_GU_VI' from data source
  `C:\Users\sabus\localeds\Git\MEDS\eds223\EDS223-HW1\data\ejscreen\EJSCREEN_2023_BG_StatePc
  using driver `OpenFileGDB'
Simple feature collection with 243021 features and 223 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -19951910 ymin: -1617130 xmax: 16259830 ymax: 11554350
Projected CRS: WGS 84 / Pseudo-Mercator
```

For the analysis, only data within Philadelphia County is needed.

```r
# Filter to Pennsylvania
pennsylvania <- ejscreen |>
    dplyr::filter(ST_ABBREV == "PA")

# Filter to Philadelphia County
philadelphia <- ejscreen |>
    dplyr::filter(CNTY_NAME %in% c("Philadelphia County"))
```

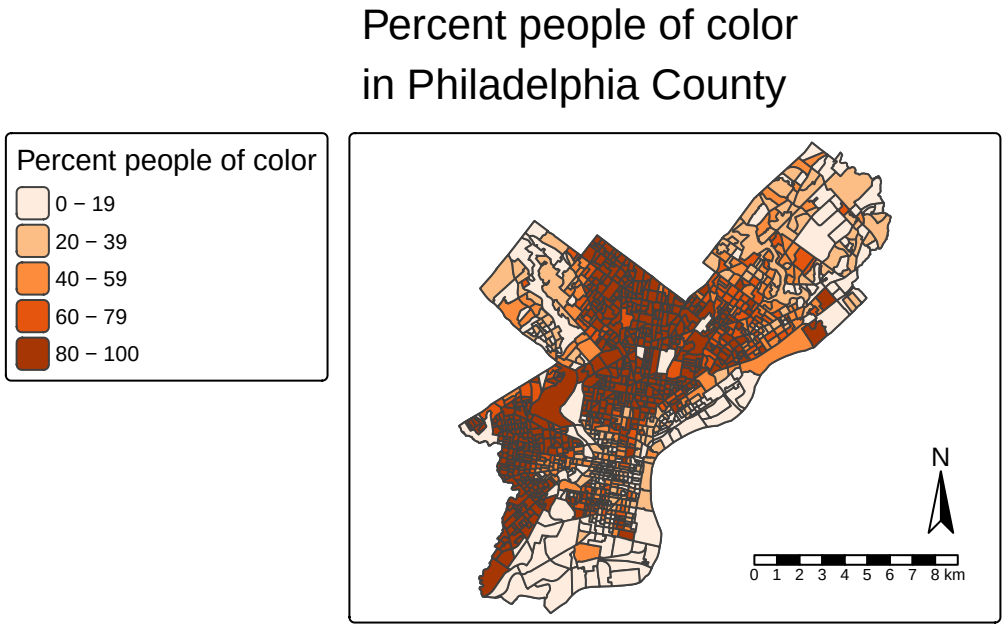
Visualize Data

To visualize correlation, two maps will be made: one mapping the distribution of people of color, and one mapping the percentage of housing units built before 1960, a proxy for potential lead exposure since older homes are more likely to contain lead paint.

```r
# Convert percent people of color from decimal to percentage
pct_ppl_color <- philadelphia['PEOPCOLORPCT'] |>
    mutate(PEOPCOLORPCT = PEOPCOLORPCT * 100)
```

```r
# Plot percent people of color at census block level
pct_ppl_color_map <- tm_layout(asp = 0, inner.margins = 0.02) +
    tm_shape(philadelphia['Shape']) +
        tm_borders() +
    tm_shape(pct_ppl_color) +
        tm_polygons(
            fill = 'PEOPCOLORPCT',
            fill.scale = tm_scale(values = 'Oranges'),
            fill.legend = tm_legend(title = "Percent people of color", position = tm_pos_out
            ) +
    tm_title("Percent people of color \nin Philadelphia County") +
    tm_compass() +
    tm_scalebar()
pct_ppl_color_map
```

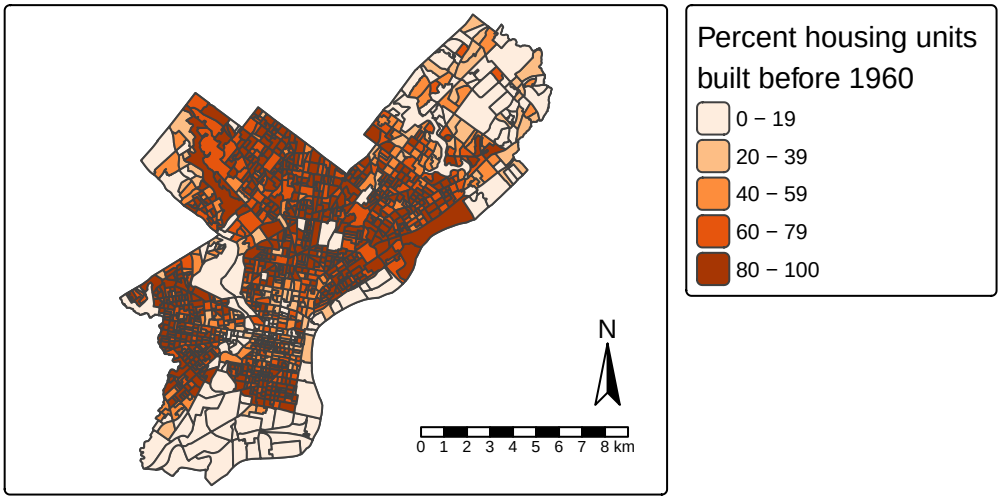
```
[cols4all] color palettes: use palettes from the R package cols4all. Run
`cols4all::c4a_gui()` to explore them. The old palette name "Oranges" is named
"brewer.oranges"
Multiple palettes called "oranges" found: "brewer.oranges", "matplotlib.oranges". The first
```




```
# Plot percent housing units built before 1960 (lead risk)
pct_pb_pnt_map <- tm_layout(asp = 0, inner.margins = 0.02) +
    tm_shape(philadelphia['Shape']) +
        tm_borders() +
    tm_shape(philadelphia['P_LDPNT']) +
        tm_polygons(
            fill = 'P_LDPNT',
            fill.scale = tm_scale(values = 'Oranges'),
            fill.legend = tm_legend(title = "Percent housing units \nbuilt before 1960", posi
            ) +
    tm_title("Percent of housing units built \nbefore 1968 in Philadelphia County") +
    tm_compass() +
    tm_scalebar()
pct_pb_pnt_map
```

```
[cols4all] color palettes: use palettes from the R package cols4all. Run
`cols4all::c4a_gui()` to explore them. The old palette name "Oranges" is named
"brewer.oranges"
Multiple palettes called "oranges" found: "brewer.oranges", "matplotlib.oranges". The first c
```

Percent of housing units built
before 1968 in Philadelphia County



Map interpretation

From these two maps, one can see that the percentage of housing units built prior to 1960 and the percentage of people of color overlap very similarly in many Philadelphia neighborhoods. It is important to note that the data taken from EJScreen uses the percentage of pre-1960 houses as a proxy for lead exposure risks. It does not necessarily mean lead exposure risks compare exactly to this number. Factors not taken into account include homes that have been remediated, or additional legacy environmental contamination. Still, the maps reflect how one factor of lead paint risk correlates with people of color. This aligns with studies suggesting Black toddlers are most affected by high blood lead levels in Philadelphia (Schmidt, 2025). The disproportionate affect on Black residents and other people of color in part may come from the historic redlining of the city. Further analysis using the EJScreen could include comparing lead paint risk to other socioeconomic groups including children under the age of 5 and low-income residents.

References

Baillie, K. U. (2022, April 12). Lead toxicity risk factors in Philadelphia. Penn Today. https://penntoday.upenn.edu/news/lead-toxicity-risk-factors-philadelphia

Schmidt, S. (2025, February 13). Philly children are exposed to dangerous lead more often than in other cities, analysis finds. WHYY. https://whyy.org/articles/philadelphia-children-lead-exposure/

Data Citation

United States Environmental Protection Agency. 2025. EJSCREEN. www.epa.gov/ejscreen.
Link: https://drive.google.com/file/d/1nG6Nj1bXfzQFOVMO8Km3eNy4SWu1YcIQ/view?usp=sharing